Daniel Lee

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Education

Western University — London, ON
BSc Computer Science — *Sep 2023 – May 2028 | GPA: 3.5/4.0*

- **Relevant coursework:** Operating Systems, Databases, Machine Learning, Reinforcement Learning, Data Science

Experience

Software Engineering Intern — Sep 2026 – Feb 2027
INVENTIS — *Seoul, South Korea*

- Traced 36 REST API flows across the multi-layer **C#/ASP.NET** backend of a dealer management SaaS platform, and documented each as a Mermaid sequence diagram to map end-to-end request flow for the team.
- Configured a Unity development environment (Unity Hub, VR Builder) and evaluated VR Builder's tooling for upcoming AR/VR product work.

Full-Stack Developer — May 2025 – Aug 2025
Kingsbridge Rehab Centre — *Mississauga, ON*

- Built and shipped the clinic's web application end to end with **Next.js 15**, **React 19**, and TypeScript, for a practice handling **1,000+** patient visits.
- Developed a timezone-aware availability module with date-fns that computes open and closed status across the 7-day schedule, eliminating manual updates.

Projects

ROCSim | *C++17, CUDA, HIP/ROCm, Qt 6, OpenGL*

- Wrote CUDA and HIP/ROCm compute kernels for a real-time N-body and collision physics engine that runs on both NVIDIA (RTX 3070) and AMD (RX 6800) GPUs, holding **60 FPS** at 500–1,000 particles.
- Implemented 3 physics models (gravity, collision, spring) and 2 integrators (Verlet, RK4) behind Strategy and Factory patterns, allowing runtime model switching with adaptive timesteps.
- Built a Qt 6.5/OpenGL interface that renders up to 5,000 particles through shared GPU buffers, with live controls, energy-based color coding, and an FPS readout.

Sentimentrix | *Python, BERT, scikit-learn, Jupyter*

- Built a Reddit sentiment pipeline that labels **150,000** comments as positive, neutral, or negative with VADER, then trained an SVM on BERT embeddings of 20,000 comments to predict those labels: **78.4%** test accuracy, 0.77 weighted F1, 0.83 ROC AUC.
- Benchmarked a linear SVM on TF-IDF (unigrams and bigrams) against an RBF SVM on PCA-reduced BERT embeddings, tuning both with GridSearchCV and 3–5 fold cross-validation.

THOTH | *React Native, TypeScript, Express.js, Supabase (PostgreSQL)*

- **Owned the entire Next.js frontend** for Thoth, a Wi-Fi motion-tracking research platform, integrating a FastAPI backend and Supabase.
- Built the chat interface for the site's AI assistant, letting visitors get answers about the project without contacting the research team.

Technical Skills

Languages: C#, C/C++, Python, TypeScript, JavaScript, SQL (PostgreSQL), Java, HTML/CSS
Frameworks and Libraries: ASP.NET, React, React Native, Next.js, Node.js, Express.js, Qt, scikit-learn, pandas, NumPy, Matplotlib, PyTorch, FastAPI
Tools and Platforms: Git, Linux, CUDA, HIP/ROCm, OpenGL, Supabase, Jupyter, Google Cloud Platform